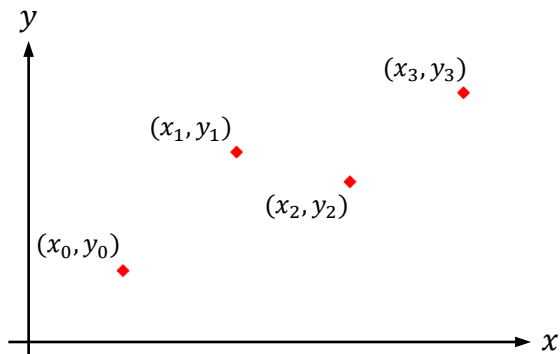


Concept & Outline



For a given set of $n + 1$ data points with $n \geq 1$, the derivative of order k can be approximated when $1 \leq k \leq n$ at a point x where $x_0 \leq x \leq x_n$.

For example, derivatives of order $k = 1, 2, 3$ can be approximated when $n = 3$ at a point x where $x_0 \leq x \leq x_3$.

⑤ Numerical Differentiation

- Approximation of Derivatives via Interpolation
Differentiation of interpolation function.
- Approximation of Derivatives by Finite Difference
Approximation from definition using finite Δx .
 - ❖ Forward Difference Formula
 - ❖ Backward Difference Formula
 - ❖ Centered Difference Formula
- ✓ ➤ The Method of Undetermined Coefficients
General method for constructing of numerical differentiation formula.
 - ❖ Formulas for Uniform Intervals
 - ❖ Formulas for Non-Uniform Intervals
 - ❖ Formulas at Arbitrary Points